9.3 Conditional Probability

Name _____ Period ____ Date _____

Algebra I 9.3 (30-minute plan) · Student sheet

I can…

- find a conditional probability from a sample space or a two-way table.
- use the formula for conditional probability.
- use conditional probability to solve real-life problems.

Vocabulary · fill in the missing word

Conditional probability — the probability that event B happens given that event A has _____, written P(B | A)

P(B | A) — read "the _____ of B given A"

Formula — P(B | A) = _____

Part 1 · Notes and examples (fill in each note, then work the example)

Conditional probability: P(B | A) means the probability of _____, given that _____ has happened.

"Given": "Given" shrinks the _____ to the outcomes in A.

Example 1

A bag holds 3 red marbles and 2 blue marbles. You choose one marble, keep it, then choose another. Given that the first marble is red, what is the probability that the second is blue?

From a table: P(B | A) = (number in A and B) ÷ _____

Example 2

A survey of 100 students asked about exercise and sleep.

a. Find P(8+ hours of sleep | exercises).

b. Find P(exercises | 8+ hours of sleep).

	8+ hours sleep	Less	Total
Exercises	36	24	60
Doesn't	14	26	40
Total	50	50	100

Formula: P(B | A) = _____ ÷ _____

Example 3

At a school, 60% of the students take Spanish, and 24% take Spanish and art. What is the probability that a student who takes Spanish also takes art?

Part 2 · Partner practice

Partner practice

Use the exercise and sleep table. Find P(doesn’t exercise | less than 8 hours of sleep).

	8+ hours sleep	Less	Total
Exercises	36	24	60
Doesn’t	14	26	40
Total	**50**	**50**	**100**